Collaborate to Elevate: Leveraging Social Capital in Digital Creator Collaborations

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1. Introduction

In the digital creator economy, followers are capital, shaping creators' visibility, monetization opportunities, and bargaining power (e.g., Labrecque et al., 2013; De Veirman, Cauberghe, and Hudders, 2017; Cheng and Zhang, 2024). As this market expands — it is projected to reach $480 billion by 2027[1] — the competition for audiences' attention has intensified (e.g., Iyer and Katona, 2015; Rossi and Rubera, 2021). Despite over 200 million active creators worldwide, a mere 20% achieve followers exceeding 10,000, and only 12% of full-time creators earn more than $50,000 annually.[2] To gain more followers, creators need to increase awareness and differentiate themselves. Recently, many creators have sought to accomplish these goals through strategic collaborations, producing content jointly with peers.[3] For instance, when Marques Brownlee, a creator who reviews technical products, partnered with Peter McKinnon, a creator specialized in photography and cinematography, for a camera review video (see Figure 1a), their collaborative video attracted over 22.8 million impressions, substantially outperforming either creator's individual content. Major social media platforms, like Instagram, TikTok, and Douyin (Chinese TikTok), recognizing the rise in creator collaborations, have introduced native features intended to facilitate the collaborative process. This trend has even catalyzed the development of specialized creator networks like *Kit.com*, which connect creators and recommend optimal collaboration partners.

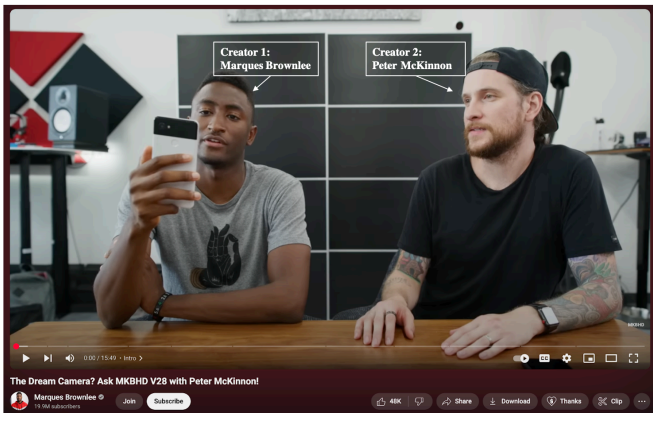


Figure 1. An Example of Video Collaboration

[1] https://www.goldmansachs.com/insights/articles/the-creator-economy-could-approach-half-a-trillion-dollars-by-2027

[2] https://linktr.ee/creator-report/

[3]https://www.youtube.com/watch?v=dAtGDfq2xjE

Despite its popularity on social media, the dynamics of creator collaboration remain largely underexplored in the academic literature. While collaboration has been widely studied in traditional media domains, including brand marketing (Rao et al., 1999; Rodrigues et al., 2011), film production (Cattani and Ferriani, 2008), and music (Uzzi and Spiro, 2005; Deshmane and Martínez-de-Albéniz, 2023), collaboration between content creators on social media is a distinct phenomenon, in several respects.

First, audiences play a uniquely central role on social media. In traditional media, communication flows in one direction, and audience feedback is limited and delayed (Lee and Tandoc, 2017). Social media changes this dynamic by enabling real-time audience interaction through likes, comments, and shares, which give creators immediate and granular signals about how their content is received (Song and Ho, 2024). Social media also amplifies the role of audiences by distributing collaborative content to the audiences of all participating creators through recommendation algorithms and built-in collaboration features. Traditional media does not offer this kind of direct, cross-audience exposure. Therefore, the composition and preferences of the creator's audience naturally become an important factor in collaboration decisions on social media.

Second, social media enables content production at a lower cost and higher frequency, compared to traditional media. Creators can easily produce content leveraging platforms' tools and features, which make it straightforward to incorporate music sampling, video filters, and templated editing features (Kaye et al., 2021). Further, because platforms' algorithmic feeds prioritize recency and consistency, creators are required to post frequently to maintain visibility.[4] The combination of low production costs and strong algorithmic incentives creates a highly fluid environment in which creators must initiate, execute, and refine collaborations at a pace far surpassing that of traditional media, where collaboration cycles often span months or even years.

The above suggests that creators face important fundamental questions. Most obviously: Who should they partner with? How frequently should they collaborate? We seek to address those questions here by i) investigating ideal collaboration strategies and ii) exploring ideal collaboration frequency and broader content production dynamics that can enhance follower growth for social media creators.

To explore the first question, we examine the effects of collaboration on follower counts, considering heterogeneity in outcomes depending on collaboration partners' characteristics. Prior research highlights that artists generally select collaborators on the basis of various characteristics, including economic status, social connections, and personal style (e.g., Cao and Soresco, 2013; Deshmane and Martínez-de-Albéniz, 2023). In social media, however, because audiences play a more active role and are not simply passive consumers, creators also need to consider the nature of a potential collaborator's

[4]https://ads.tiktok.com/business/en-US/blog/tiktok-creative-ultimate-guide?redirected=1&acq_banner_version=73758464

follower base, particularly the degree to which it aligns or diverges from their own, a concept we refer to as ‘audience divergence’.

Whether creators stand to gain more by partnering with others whose audiences resemble their own or with those whose audiences are quite different is a source of some debate in the practitioner community.[5] That debate reflects a theoretical tension in the academic literature as well. Drawing on theories of bonding and bridging social capital (Putnam, 2002), we argue that collaborating with creators who attract similar audiences reflects an investment in bonding social capital, which has the potential to reinforce existing network connections and facilitate trust (Granovetter, 1973). In contrast, collaborating with those who maintain divergent audiences represents an investment in bridging social capital, which has the potential to provide creators with access to new and different communities (Paxton, 2002). Here, by theorizing audience divergence as a strategic determinant of collaboration outcomes, we extend existing academic work on partner selection and highlight a mechanism uniquely salient in the social media environment.

To address our second question, related to collaboration frequency and dynamics, we empirically examine creators’ content production strategies after a collaboration. In doing so, we consider that creators face choices over whether to return to solo content, engage in another collaboration, or pause releasing new content temporarily. Each option carries its own advantages: solo content reinforces a creator’s individual identity with new audiences; another collaboration can extend the reach of the previous one by exposing to additional partner audiences; and a brief pause can avoid oversaturating followers and allow audience attention to reset (Ravindran et al., 2014; Deshmane and Barriola, 2024). These choices embed notions not only of the optimal frequency of collaboration but also the broader dynamics of content production in a high-velocity environment. By studying these patterns, we provide insights into how creators can optimally balance solo output, collaborations, and breaks, to sustain their audience growth.

We undertake our analyses leveraging data from Douyin, China’s leading short-video social media platform, with more than 750 million monthly active users as of March 2024.[6] Creator collaborations on Douyin occur through a built-in collaborative feature, where two creators jointly conceptualize, negotiate, and produce a collaborative video that appears on both profiles. We identify a list of collaborative videos and their respective creators, and associated metadata. Our dataset covers collaborative and solo videos that span five subject matter categories—Life Sharing, Medical Health, Finance, Technology, and Science—posted during March and April of 2024. For each video, we collect the creator’s follower counts, other video posting behaviors, video engagement metrics, and collaboration

[5]https://www.thetilt.com/audience/creator-collaboration
[6] https://www.statista.com/statistics/1361354/china-monthly-active-users-of-douyin-chinese-tiktok/

frequency at multiple time points surrounding the focal video's posting. We also gather creator-level audience profiles that summarize compositions of each creator's follower base across several demographics, such as gender, age, and geographic locations, and behavioral attributes, such as subject-matter interest, purchasing power, and socioeconomic segments.

When estimating the impact of collaboration on a creator's follower counts, we address potential confounding factors that may jointly influence collaboration decisions and the outcome via propensity score matching, or PSM (Rosenbaum and Rubin, 1985). Each collaborative creator-video is matched to one similar non-collaborative creator-videos based on several matching covariates, including the creator's recent video posting activities, characteristics (e.g., audience size, average engagement, collaboration frequency, etc.), and audience compositions (e.g., interest, city tier, age, and gender). Further, we match on 16-dimensional embeddings of video content embeddings based on videos' textual transcripts. The embeddings, obtained from OpenAI's text-embedding model (Neelakantan et al., 2022), provide a high-fidelity representation of content, enabling us to isolate the role of partner selection and post-collaboration content strategy in our analyses.

To address our first question on partner selection, we quantify differences in creators' audience characteristics as the Jensen-Shannon (JS) distance between collaborators' audience profiles (Lin, 1991). This measure allows us to understand whether follower growth deriving from collaborative videos is greater when collaborators' audiences are more or less similar. To address our second question on collaboration frequency and content dynamics, we construct a categorical variable capturing the creator's actions following a collaboration: posting solo, engaging in another collaboration, or taking a break from releasing new content. This variable reveals how different after-collaboration strategies shape the collaboration outcomes and shed light on the optimal frequency of collaboration and content production.

Our results reveal that posting a collaborative video (versus an equivalent solo video) increases a creator's follower growth rate (i.e., differences in log(followers)) by an average of 0.16% ($p<0.01$) over the first few days following video posting. Further, the effect is significantly larger when the collaborator's audience differs from the focal creator's audience in terms of subject-matter interests ($p < 0.05$). We observe a similar pattern for divergence in socioeconomic segments, where greater differences in followers' socioeconomic backgrounds likewise amplify the collaborative benefit ($p < 0.05$). These findings highlight that collaborations yield higher returns when they bridge across distinct audience groups, supporting the value of bridging social capital in audience expansion. Finally, our analyses of after-collaboration content creation strategies demonstrate that creators benefit most from a collaboration when they post solo content afterwards (rather than engaging in another collaboration or remaining inactive in posting) ($p < 0.05$).

We evaluate the robustness of our findings in several ways. First, to validate the comparability of matched treatment and control groups, we estimate an event study specification to examine the pre-treatment trends in follower growth. The absence of differential trends between groups supports the validity of our matching procedure and mitigates concerns about reverse causality, specifically, the possibility that creators experiencing slower growth selectively pursue collaborations to boost visibility. Second, we test the sensitivity of our results to alternative outcome windows by re-estimating effects using follower growth measured at varying points in time after collaboration.[7] The results remain consistent across these specifications.

Our research extends the collaboration literature to the context of social media content creators. We reframe partner selection strategy within this highly interactive and dynamic environment. While prior studies often examined similarity or difference in collaborator attributes such as brand image, social status, or network position (e.g., Cao and Sorescu, 2013; Deshmane and Martínez-de-Albéniz, 2023), we emphasize the role of collaborators' audience divergence and assess how this divergence moderates the impact of collaboration on creators' follower growth. In addition, we apply social capital theory to examine collaborator selection for social media creators, expanding its application beyond traditional settings (Coleman, 1988; Putnam, 2002). Building on this theoretical foundation, we develop a conceptual framework for understanding the tension creators face when choosing partners whose audiences differ (or not) from their own. This paper also contributes to the literature of influencer marketing by investigating how joint influencer efforts shape marketing outcomes. While previous studies have primarily focused on individual influencer actions (e.g., Leung et al., 2022; Zheng and Zhang, 2024), our study addresses this gap by exploring how content collaboration between influencers impacts their follower counts, providing new insights into collaborative influencer marketing.

These findings also offer clear practical guidance for social media creators who seek to grow followers through content collaboration, and for platforms that attempt to support them. For creators, our results suggest that collaboration is more beneficial when collaborators have different audience compositions, especially in behavioral attributes such as interests, purchasing power, and socioeconomic status. Collaborating with someone whose followers differ meaningfully from one's own can facilitate bridging social capital, connecting to new audiences, and enhancing follower growth.

For platforms (e.g., Creator Network by Kit.com, TikTok Creator Marketplace) and talent agencies (e.g., multi-channel network (MCN) companies), our findings highlight the value of incorporating potential collaborators' audience compositions, rather than solely their self-attributes, into collaborator-matching algorithms. Doing so can improve the quality of recommendations and better support creators to expand their follower base through content collaboration on social media.

[7] Follower growth measured at one, three, four, and five days after. the video post.

2. Related Literature

2.1 Social Media Creator Audiences

Firms are increasingly turning to digital creators to promote their brands and products on social media (Liang et al., 2022). In selecting from millions of potential partners, followership (follower count) often emerges as a key criterion. Unlike celebrities who typically attract fans through institutionalized media exposure, e.g., television, film, and music (McQuarrie, Miller, and Phillips, 2013), social media creators accumulate followers post by post through curated content and personalized interactions (Luo and Yuan, 2019). Follower count has been widely used as a proxy for a creator's popularity (De Veirman, Cauberghe, and Hudders, 2017), reputation (Cheng and Zhang, 2024), and perceived credibility (Labrecque et al., 2013). Moreover, follower count captures an enduring form of audience engagement, signaling the audience's sustained interest and willingness to maintain an ongoing relationship with the creator (Brodie et al., 2011; Morton, 2020).

A growing body of research has identified various factors that influence creators' follower levels (e.g, Caro and Martínez-de-Albéniz, 2020; Cheng and Zhang, 2024; Nistor, Selove, and Villas-Boas, 2024; Song and Ho, 2024). For instance, Cheng and Zhang (2024) show that posting sponsored content can lead to a loss of followers, underscoring the delicate balance between monetization and audience trust. Nistor, Selove, and Villas-Boas (2024) find that being authentic helps early-stage creators to build awareness quickly and maximize follower acquisition. Existing studies primarily examine the impact of individual-level actions on creators' follower counts, yet less is known about how collaborative endeavors influence follower counts. We address this gap by exploring how content collaboration between creators impacts their follower counts.

2.2 Collaboration and Partner Selection

Collaboration has been widely explored in various forms, from brand alliances (e.g., Rao et al., 1999; Cao and Soresco, 2013) to film production (e.g., Cattani and Ferrini, 2008; Gibson and Dibble, 2012), music (Uzzi and Spiro, 2005; Deshmane and Martínez-de-Albéniz, 2023), and, more recently, content production on social media (e.g., Koch et al., 2018; Lode et al., 2018; Yang et al., 2023). The existing literature presents mixed evidence regarding the impact of collaboration. On the one hand, collaboration is frequently viewed as a means to expand reach (Rodrigues et al., 2011; Deshmane and Martínez-de-Albéniz, 2023), increase the visibility of the involved parties (Rao et al., 1999; Koch et al., 2018), and create fresh content (Yang et al., 2023). On the other hand, collaborations introduce challenges, necessitating greater communication and imposing coordination demands (Porac et al., 2004; Cummings and Kiesler, 2007), with potential issues arising around fairness and equality of participation, e.g., social loafing, free riding (Latané et al., 1979; Albanese and Van Fleet, 1985), which may hinder

collaborative success (Keller and Lane, 2003; Uzzi and Spiro, 2005). These divergent findings highlight that the outcomes of collaborations are highly context-dependent. This further emphasizes the need to understand collaborations in the specific context of social media creators. Our paper builds on and extends emerging research on content collaboration in social media by examining how partner selection strategies may shape the success of collaboration in this setting.

Partner selection has long been recognized as a key determinant of collaboration success. However, existing studies offer conflicting views on whether similarities or differences between collaborators lead to better outcomes. Some research emphasizes the value of consistency between partners. For example, Simon and Ruth (1998) find that co-branded products generate more favorable consumer responses when the collaborating brand identities are aligned. Similarly, Cao and Soresco (2013) show that brand collaborations with different brand images improve financial market reactions. In contrast, other studies highlight the advantages of differentiation between partners. Deshmane and Martínez-de-Albéniz (2023) discover that musical collaborations between artists of different status levels and social networks may result in greater streaming success. A third perspective suggests that moderate differences between partners can produce the best collaborative output (e.g., Uzzi and Spiro, 2005; Cattani and Ferriani, 2008; Gibson and Dibble, 2012). Notably, however, these studies focus primarily on collaborator self-attributes, such as brand identity, social status, or network position; they give limited attention to the ultimate consumers of collaboration output. This emphasis in prior literature on creator attributes is natural, as it reflects traditional media structure, where communication is typically unidirectional and audiences exert minimal influence on production decisions (Lasswell, 1948). In such contexts, partner selection is naturally driven by internal synergy between collaborators themselves, rather than by any consideration of their respective audiences.

Creator collaborations on social media, however, unfold under different conditions. Social media fosters a bi-directional communication environment in which creators interact directly with their audience, receive real-time feedback, and adapt content strategies dynamically (Song and Ho, 2024). Unlike traditional media, where producers broadcast to passive viewers, social media creators cultivate bi-directional and ongoing relationships with followers, who may therefore exert a strong influence over the selection of collaboration partners (Hu et al., 2020). Moreover, collaborations on social media are distributed not only to the creator's own followers but also across collaborators' audiences through recommendation algorithms and built-in collaboration features, creating direct cross-audience exposure. This interactive and cross-audience dynamic introduces a new dimension in partner selection: beyond the potential collaborators themselves, one must also consider who their audiences comprise. We thus extend the collaboration literature, introducing audience divergence as a new strategic consideration in partner selection. Specifically, we define audience divergence as the degree of difference between collaborators'

follower composition in a specific dimension (e.g., interest, demographics, etc.). Understanding how audience divergence influences the impact of collaboration on a creator's follower counts is critical for identifying effective partner selection in the social media context.

2.3 Social Capital Theory

Social capital is defined as "the sum of actual and potential resources embedded within, available through, and derived from the network of relationships possessed by that individual" (Nahapiet and Ghoshal, 1998, p. 243). Social capital theory emphasizes how such relational resources can be mobilized for individual or collective benefits (Coleman, 1988; Nahapiet and Ghoshal, 1998). In digital settings, social capital theory has been applied to explain relational values generated through producer-producer ties, such as relationships between musicians (Deshmane and Martínez-de-Albéniz, 2023), online community contributors (Forderer and Burtch, 2025), and producers and consumers, such as connections between IT service providers and users (Sun et al., 2012). Building on the producer-consumer perspective, we conceptualize a creator's follower base as part of their social capital, since it captures not only the ongoing relationship between creators and their audiences but also the attention, trust, and influence embedded in these ties—resources that can translate into monetization opportunities (Goldenberg et al., 2019). In collaborations, creators mobilize their social capital by providing each other with potential exposure to their respective follower communities.

2.3.1 Bonding Social Capital

Social capital is often categorized in two complementary dimensions: bonding and bridging (Putnam, 2002). Bonding social capital refers to the value embedded in tight-knit relationships within homogeneous groups, such as those sharing common values, interests, or backgrounds. Putnam (2002) refers to it as the 'sociological superglue' that reinforces group identity, fosters belonging, and strengthens internal connections. This form of social capital is characterized by high trust and emotional intimacy (Granovetter, 1973), which promotes efficient information exchange and cooperation (Adler & Kwon, 2002). However, its cohesive nature can limit network diversity, thereby heightening the risk of groupthink (Janis, 1982) and social exclusivity (Sajuria et al., 2014).

In creator collaborations, when a collaborator's audience composition is similar to that of a focal creator's audience, the collaboration can be viewed as an investment in bonding capital. This is because collaborating with creators who maintain similar audiences, whether in terms of interests, demographics, or socioeconomic backgrounds, reinforces audience consistency and deepens resonance with existing followers. This form of collaboration may be beneficial because familiarity and trust can facilitate smoother integration and higher follower conversion. However, homogeneity in collaborators' audiences

may also constrain the collaboration's follower growth potential, as the opportunities to access new audience segments may be limited.

2.3.2 Bridging Social Capital

Bridging social capital, by contrast, arises from relationships that bridge heterogeneous demographic, socio-economic, or interest-based divides (Putnam, 2002). Unlike bonding social capital that strengthens existing networks, bridging social capital acts as sociological leverage, connecting different communities and facilitating the exchange of ideas and feedback (Burt, 1992; William, 2006, p. 597). It also promotes greater tolerance and open-mindedness toward different people and ideas (Paxton, 2002). Yet, bridging social ties often lack the emotional depth and stability found in bonding relationships, making them more fragile and transactional in nature (Putnam, 2002).

In creator collaborations where a collaborator's audience diverges from that of a focal creator, the collaboration can be viewed as an investment in bridging social capital. When creators partner with others whose audiences differ substantially in demographics, socioeconomic backgrounds, or interests, they connect to a broader audience. Bridging social capital fosters the exchange of diverse information and encourages open-mindedness. The novelty of encountering unfamiliar creators can spark curiosity and cognitive engagement, increasing the likelihood of exploration and eventual subscription (White, 1969, p.317). However, such collaborations also carry higher relational risk: the lack of familiarity and alignment between creators and their partner's audience can lead to weaker emotional connections and lower receptiveness.

Overall, given this natural tension, it is unclear how audience divergence moderates the benefits of collaboration for social media creators, highlighting the need for further empirical investigation. By integrating social capital theory with the concept of audience divergence in creator collaborations, we provide a framework for understanding how varying levels of audience divergence influence value creation through collaborations.

3. Methods

3.1 Study Context

Our analysis focuses on Douyin (also known as Chinese TikTok), the leading short-video social media platform in China. In August 2022, Douyin introduced a built-in co-creation feature (see Figure 2) to facilitate content collaboration between creators. Videos published through this feature are automatically displayed under a dedicated "collaboration" tab, which highlights the participating creators (including their names, photos, and profile links), and are shared with followers of both contributors. Engagement metrics (e.g., likes, comments, shares, etc) are aggregated and displayed on each participant's profile.

Unlike prior studies that require machine learning tools to infer collaboration (e.g., Koch et al., 2018; Lode et al., 2018; Chen et al., 2020), this platform-level feature allows us to readily identify collaborative videos. As such, Douyin presents a unique empirical setting to examine the impact of collaboration on creators' followership and how that impact varies with partner (and audience) characteristics, as well as post-collaboration content creation choices.



Figure 2. A Snapshot of Video Collaboration Feature on Douyin

3.2 Data

Our unit of analysis is the creator-video; that is, each observation corresponds to a video posted by a specific creator. We identify all co-created videos and their respective creators, posted during March and April 2024, across five subject matter categories: Life Sharing, Medical Health, Finance, Technology, and Science. Although collaborations can involve several creators, for simplicity, we focus on collaborations involving just two, which account for 85% of all collaborative content in our study period. To benchmark the effects of collaboration, we also randomly sample solo videos from the same five content categories during the same time window, along with their respective creators. After removing observations where creator characteristics, audience characteristics, or video characteristics were missing, we arrive at a final sample of 13,287 unique creator-video observations. This sample includes 2,093 collaborative creator-video instances with 1,441 collaboration videos and 1,355 creators, and 11,194 solo creator-video

instances with 11,194 solo videos and 1,786 creators. Table 1 describes key variables used in the study; we discuss them in greater detail below.

3.2.1 Creator Characteristics

For each creator, we gather daily follower counts over 30 days preceding and following the focal video posting date. We categorize creators based on their reach, applying the influencer taxonomy proposed by Sonnenberg (2021). Thus, we classify creators as Nano (≤1K), Micro (1K–100K), Macro (100K–1M), or Mega (>1M), based on their follower counts the day before video posting. These tiers reflect a creator's differential market influence and credibility (Labrecque et al., 2013; Goldenberg et al., 2019), and thus have the potential to moderate audience engagement (Wiles et al., 2022). Additionally, we measure creator tenure as the number of days since the account's creation on Douyin. Longer tenure reflects greater familiarity with content production norms and audience dynamics on Douyin (Cheng and Zhang, 2024).

We also collect each creator's video posting history, prior to the focal video, including timestamps, video type (solo or collaborative), and engagement metrics such as likes, comments, shares, and saves. From these records, we calculate the prior collaboration frequency of creators, defined as the proportion of collaborative videos among all preceding video uploads[8]. This metric reflects the creator's previous experiences with collaborations, allowing us to distinguish between habitual collaborators and those for whom collaboration is an infrequent or novel strategy. Prior collaboration experience is likely to influence not only the creator's decision to engage in collaboration but also its effectiveness. Finally, we track the number of solo and collaborative videos published during each of the five days following the focal post, helping us to capture after-collaboration behavior for each creator.

[8] Due to platform's data retrieval restriction, we are only able to gather up to 100 videos (if a creator posted more than 100 videos) prior to the focal video.

Table 1. Variables.

Variable	Descriptions	Mean	SD	Min.	Max.
$FollowersBefore_{ij}$	Creator *i*'s follower counts 1 day before video *j*.	1,324,904	4,181,713	1,138	70,534,664
$FollowersAfter_{ij}$	Creator *i*'s follower counts 1 day after video *j*.	1,332,423	4,199,800	1,137	70,636,292
$CareerLength_{ij}$	Number of days from creator *i* joined Douyin to video *j*.	1,620	659.198	11	2,745
$AverageEngagementBefore_{ij}$	Average number of interactions (likes + comments + shares + saves) for the videos posted during prior 5 days by the creator *i* to video *j*.	53,147	166,833.1	0	3,930,116
$CollaborationFrequencyBefore_{ij}$	Creator *i*'s ratio of collaborative videos before video *j*.	0.0217	0.0975	0	1
$SoloVideoCountBefore_{ij}$	Averge number of solo videos posted by creator *i* during five days before video *j*.	3.255	2.556	0	42
$CollaborativeVideoCount_{ij}$	Average number of collaborative videos posted by creator *i* during five days before video *j*.	0.141	0.454	0	6
$PostingDayWeekend_{ij}$	1 if creator *i* posted video *j* on weekend.	0.282	0.450	0	1
$PostingTimeMorning_{ij}$	1 if creator *i* posted video *j* in the morning.	0.148	0.355	0	1
$PostingTimeEvening_{ij}$	1 if creator *i* posted video *j* in the evening.	0.600	0.490	0	1
$PostingTimeMidnight_{ij}$	1 if creator *i* posted video *j* in the midnight.	0.0498	0.217	0	1
$Number\ of\ videos\ posted_{ij}$	Number of videos creator *i* posted on the same day as video *j*.	1.121	1.094	1	100

To capture a creator's audience engagement level at the time of the focal video, we compute the average engagement rate based on videos posted during the five days before the focal video. For each

preceding video, the total engagement is measured as the sum of likes, comments, shares, and collects. Averaging these yields a creator-video level proxy for content resonance and audience responsiveness at the time of posting the collaboration (Swaminathan, Hughes, & Brooks, 2019), which may in turn affect the impact of collaboration on follower counts.

3.2.2 Audience Characteristics

Each creator is linked to a detailed audience profile, which captures the composition of their follower base. Douyin first profiles each individual user across several demographic and behavioral attributes. Demographic attributes, including gender, age, and geographic location (e.g., city tier of residence), are drawn directly from information reported on the user's account page. Behavioral attributes, including subject-matter interests, purchasing power, and platform-defined consumer segments, are revealed through users' online activities, such as their browsing, interacting, and purchasing behaviors (Trusov et al., 2016).

Douyin then aggregates these user-level attributes to produce creator-level audience profiles. Each profile is represented as a probability distribution that reports the proportion of followers falling into each predefined category. For example, the gender profile shows the share of male and female followers, the age profile reports the percentage of followers in five age brackets (18–23, 24–30, 31–40, 40–50, and >50), and the city-tier profile shows the distribution of followers across eight tiers of China's urban classification system, derived from their geographic locations.

Behavioral attributes are summarized using the same approach. The interest profile covers 20 topical categories (e.g., Life Sharing, Science, Sports, etc.) based on videos that users consistently watch, interact with, or search for; it captures the percentage of followers engaged with each interest topic. The socioeconomic-segment profile is an eight-category distribution, indicating the proportion of followers in each platform-defined socioeconomic group (urban middle class, urban white-collar, urban blue-collar, urban senior, rural senior, rural junior, Generation Z, and stylish mom).

Together, these distributions provide a precise description of each creator's audience composition. Creators can view these statistics directly in Douyin's creator tool. See Appendix 1 for summary statistics of all audience profiles used in our analysis.

3.2.3 Video Characteristics

For each video, we collect key metadata, including posting timestamp, engagement metrics, and the number of other videos posted by the same creator on the same day. Using the timestamp, we construct indicators capturing aspects of posting time (weekday/weekend; time of day) to account for variations in audience activity on the Douyin platform.

To address the potential confounding influence of video content—perhaps collaboration videos include systematically different content than solo videos—we construct measures of the semantic subject matter of each video. We use the spoken transcript of each video to proxy for this, transcribing the audio using OpenAI's Whisper Large model (Radford et al., 2022), a state-of-the-art automatic speech recognition system trained on extensive multilingual and multitask supervised data. Its documented strong performance in Mandarin ensures transcription quality in our setting.[9] Following transcription, we convert the resulting text into a 16-dimensional embedding using OpenAI's large pre-trained text embedding model (Neelakantan et al., 2022). This model was trained using Matryoshka Representation Learning (MRL), which encodes information at multiple levels of granularity and allows dimensionality reduction without major content loss (Kushpati et al., 2022). The resultant embeddings represent each video transcript in a continuous vector space, such that videos with similar content or messaging are mapped to nearby points in that space. By incorporating them as covariates in our matching and empirical models, we adjust for underlying differences in video content that might otherwise confound decisions to collaborate and the effects of collaboration.

4. Empirical Framework

4.1 Matching Strategy

In our research context, where creators alternate irregularly between collaborative and solo videos, it is difficult to define a common pre- and post-collaboration period based on calendar time. Instead, we treat each creator-video pair as a separate event, following Cheng and Zhang (2024), taking the time of focal video posting as the focal period of interest, aligning videos in time, and constructing measures of followership and activity before and after each. We assign creator-video pairs to the treatment group when a creator posts a collaborative video, and to the control group when a creator posts a solo video.

A key empirical challenge in our setting stems from the self-selection into collaboration. Specifically, the decision to engage in a collaborative video is not randomly assigned; rather, it is influenced by the creator, audience, and video characteristics, implying potential endogeneity problems (Athey and Imbens, 2006). To address this issue, we employ a propensity score matching (PSM) strategy (Rosenbaum and Rubin, 1985). Each collaborative creator-video observation is matched to one solo creator-video posted on the same day, using a nearest-neighbor approach based on a logit propensity score. The propensity score is estimated using a rich set of covariates that jointly predict the likelihood of collaboration, as we describe next. First, certain types of creators may be inherently more inclined to collaborate. For example, smaller-sized creators may proactively seek collaboration opportunities to

[9] When evaluated using FLEURS—a benchmark for multilingual speech tasks—Whisper achieves a word error rate of under 10% for Mandarin (Radford et al., 2022).

expand their reach, while more established creators with stable followings may feel less pressure to do so. At the same time, pre-collaboration video posting activity also plays a role, as creators who post frequently may face different logistical demands that may influence their ability or willingness to collaborate. Accordingly, we match on creators' characteristics, such as audience size and prior average engagement, and their video posting activities, such as the number of solo (collaborative) videos released on days before the focal event.

Additional audience characteristics also shape collaboration decisions. The bilateral relationship between creators and their audience suggests that creators may strategically adjust their content formats to match audience expectations. In some cases, a creator may perceive that their audiences have a stronger preference for collaborative content, prompting them to initiate more collaboration partnerships. Given the high dimensionality of audience profiles (54 raw variables), we construct matching covariates by applying Principal Component Analysis (PCA). PCA extracts orthogonal linear components that capture the dominant patterns in the data (Jolliffe, 2002). Retaining components that collectively explain at least 90% of the variance, we reduce the set to 26 orthogonal audience variables.

Finally, the content genre further shapes collaboration feasibility. Certain genres, like life-sharing, naturally support collaboration through shared narration or casual dialogue. In contrast, specialized content, like finance, often requires tighter scripting and domain expertise, limiting collaboration opportunities. Therefore, we match video content based on the 16-dimensional textual embeddings derived from video transcripts.

After matching, the absolute values of the standardized difference between treatment and control across all covariates do not exceed 0.1 (See Appendix 2 for more details). This aligns with the criteria proposed by Imbens and Rubin (2015), indicating that the matched treatment and control groups are statistically comparable.

4.2 Regression Specification

4.2.1 Main Effect

With the matched treatment and control groups, we estimate the impact of collaboration on follower counts, using equation (1):

$$DifferenceInLog(followers)^{k}_{ij} = \beta \cdot Collaboration_{ij} + \gamma \cdot Control_{ij} + \varepsilon_{ij} \qquad k \in [1, 2, 3] \qquad (1)$$

The primary outcome variable, $DifferenceInLog(followers)^{k}_{ij}$, is defined as the difference in the logarithm of follower levels observed one day before a video was posted, i.e., $log(FollowersBefore_{ij})$,

and the logarithm of follower levels observed k days after $log(FollowersAfter_{ij})$, for creator i posting video j. Because we employ the difference in logs, this measure approximates the percentage change in followers over a specific time interval, a common form of normalization (Goldenberg et al., 2019; Song and Ho, 2024). This method helps to rule out cross-sectional differences in follower levels that might influence baseline rates of change in audience engagement (Wies, Bleier, and Edeling, 2023). This approach also enables more meaningful comparisons across creators—for instance, a 10% growth in followers represents the same relative increase whether a creator has 1,000 or 100,000 followers.

$Collaboration_{ij}$ is our main independent variable of interest. This variable equals 1 when creator i posts a collaborative video j, and 0 when creator i posts a solo video j. The associated coefficient captures the main effect of posting a collaborative video on the follower growth rate of creator i, relative to posting an equivalent solo video. Finally, $Control_{ij}$ is our vector of matching covariates, capturing creator i's characteristics (e.g., creator i's average engagement, career length, and collaboration frequency), creator i's audience profiles (e.g., socioeconomic segments, interests, demographics, etc.), and creator i's pre-video posting activities to mitigate audience spillover reactions from prior content. We further control for video j's characteristics (e.g., content embeddings) to account for content variation that may influence audience responses.

We include fixed effects for the video's topical category to absorb systematic differences in audience engagement across content genres, as well as fixed effects for posting date and posting time (e.g., morning, afternoon, etc.) to adjust for temporal variation in audience activity on the platform. Standard errors are clustered at both the creator and video levels.

4.2.2 Audience Divergence

To empirically assess the heterogeneous effect of collaboration, depending on collaborators' audience divergence, we extend our baseline model by incorporating an interaction term $AudienceDivergence_{ij}$, as indicated in Equation (2):

$$DifferenceInLog(followers)^{k}_{ij} = \beta_1 \cdot Collaboration_{ij} + \beta_2 \cdot Collaboration_{ij} \cdot AudienceDivergence_{ij} + \gamma \cdot Controls_{ij} + \varepsilon_{ij} \qquad k \in [1, 2, 3] \qquad (2)$$

where the main effect of collaboration, coefficient β_1, captures the average effect of collaboration when audience divergence is at its mean level, and the interaction coefficient β_2 reveals how deviations from

the mean audience divergence value moderate the collaboration effect. Other covariates are included as in our main specification.

$$AudienceDivergence_{ij} = \sqrt{\frac{1}{2}D_{KL}(collaborator\ 1 || \frac{collaborator\ 1 + collaborator\ 2}{2}) + \frac{1}{2}D_{KL}(collaborator\ 2 || \frac{collaborator\ 1 + collaborator\ 2}{2})} \quad (3)$$

$AudienceDivergence_{ij}$ is operationalized as the Jensen-Shannon (JS) distance between two collaborators' audiences, along each five dimensions: segments, interest, age, gender, and living city tier (Lin, 1991). As shown in equation (3), JS divergence score is a symmetric and smoothed adaptation of the Kullback-Leibler (KL) divergence, effectively quantifying the differences between two probability distributions. For instance, a low JS divergence for audience subject matter interest distributions implies that two creators have audiences with similar topical interests. Since a value of zero in the $AudienceDivergence_{ij}$ measure never occurs in practice, we demean the JS divergence measure for our regressions.

4.2.3 Collaboration Frequency

Lastly, to explore the optimal collaboration frequency and understand how follow-up content creation activities may moderate the returns to collaboration, we construct a categorical variable $FollowUpAction_{ij}$ that tracks whether the creator *i* posted solo videos, collaborative videos, or did not post any content within three days of initial video *j*. As shown in equation (4), we extend the baseline model by interacting the treatment indicator with $FollowUpAction_{ij}$. The reference group for this specification is no video posting, so the main effect coefficient captures the collaboration impact when creators take a break for three days following the collaboration. Interaction terms reveal the differential effects of posting—whether solo or collaborative—after the effectiveness of initial collaboration. Because we are explicitly examining content creation activities within the three-day window following the collaboration, we focus on outcome windows measured three, four, and five days after the initial post.

$$DifferenceInLog(followers)^{k}_{ij} = constant + \beta_1 \cdot Collaboration_{ij} + \beta_2 \cdot Collaboration_{ij} \cdot FollowUpAction_{ij} + \gamma \cdot Controls_{ij} + \varepsilon_{ij} \qquad k \in [3, 4, 5] \quad (4)$$

5. Results

5.1 Main effect

Figure 3 presents model-free evidence of the effect of collaboration on creator follower growth. The y-axis captures differences in log(followers) before and after the video, while the x-axis reflects the number of days following a video's release. The red line represents average changes associated with collaborative videos, while the grey line represents average changes following the posting of solo videos. Since the y-axis measures differences in log(followers), the value approximates percentage changes in follower levels. Any positive effect corresponds to an increase in a creator's absolute follower level. The gap between the two cruvers indicates that posting a collaborative video, on average, generates higher follower growth than posting a solo video, particularly over the first few days following release.

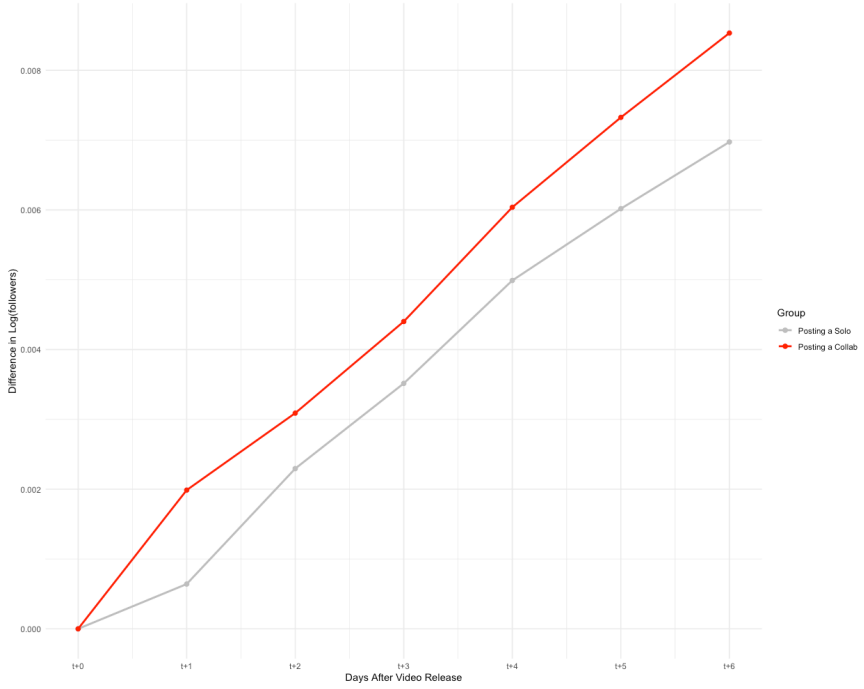


Figure 3. Model-Free Evidence of Collaboration and Difference in Log(followers)

Table 1 reports the regression estimates associated with Equation (1), quantifying the impact of collaboration on a focal creator's difference in log(followers). Consistent with the visual evidence, we find that posting a collaborative video (versus an equivalent solo video) significantly increases the follower growth rate by an average of 0.16% ($p<0.01$) when measured one day after posting. For a micro-creator with 100,000 followers, this translates to approximately 160 additional followers per collaborative video, a meaningful gain in the context of everyday content production.

The effect remains statistically positive ($p<0.1$) when measured two and three days after posting, with estimates between 0.13% and 0.15%. Although there is a slight attenuation in magnitude over time, as creators often resume posting new content within a few days, gradually diluting the initial effect, the direction and significance of the coefficients suggest that collaboration provides a sustained boost in followers beyond the initial release day.[10] This finding aligns with industry intuition and prior academic literature on collaboration, which often highlights its benefits (e.g., Rao et al., 1999; Wuchty et al., 2007; Deshmane and Martínez-de-Albéniz, 2023).

Table 1. Main Effect

	$DV: DifferenceInLog(followers)^{k}_{ij}$		
	$k = 1$	$k = 2$	$k = 3$
	(1)	(2)	(3)
Collaboration	0.0016***	0.0013*	0.0015*
	(0.0005)	(0.0007)	(0.0009)
Controls	Included	Included	Included
Fixed effects	Included	Included	Included
Observations	1,457	1,453	1450
R^2	0.1231	0.1223	0.1162

Notes. *p<0.1; **p<0.05; ***p<0.01; Estimates are rounded to four decimal digits. Clustered (at video and creator level) standard errors in parentheses.

5.2 Audience Divergence

Figure 4 presents model-free, descriptive evidence of how the impact of collaboration varies with collaborator audience divergence. The vertical axis reflects differences in log(followers). For visualization, we classify collaborations into two groups based on their JS divergence scores: collaborations with above-median JS scores are labeled as high audience divergence (red line), while those below the median are categorized as low audience divergence (blue line). The grey line depicts follower growth associated with the posting of solo content. In general, we find that collaborations involving greater audience divergence generate higher follower growth. The differences are especially pronounced along the behavioral dimensions of audience compositions: interests, purchasing power, and socioeconomic segments.

[10] In our dataset, more than 67% of creators posted new content during three days after initial video.

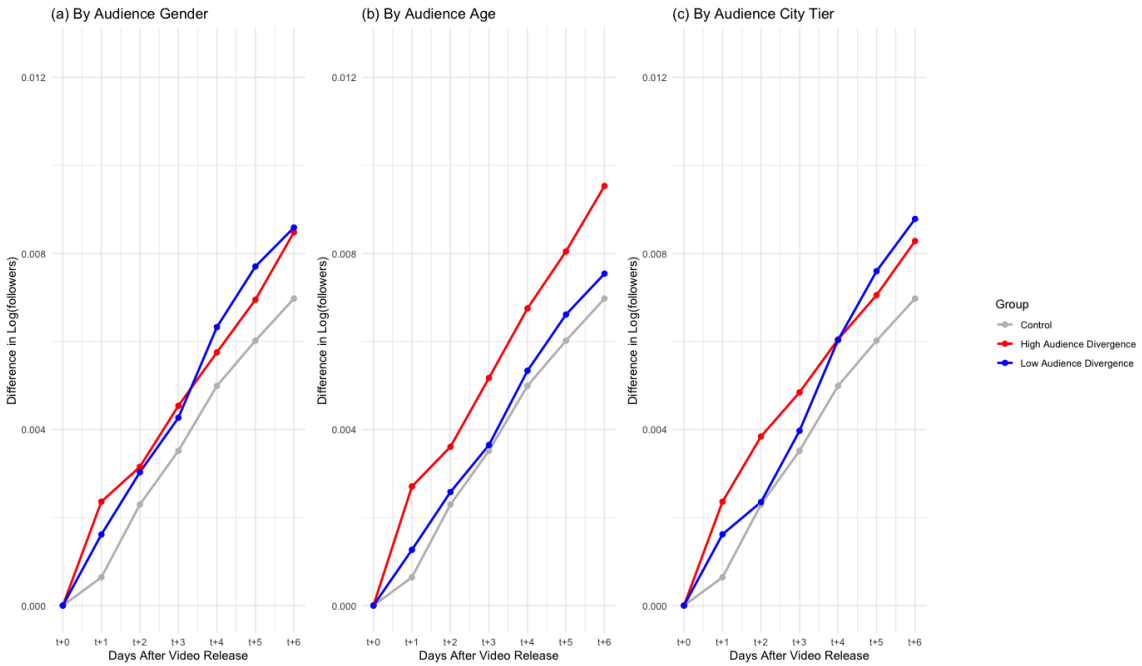


Figure 4a. Model-Free Evidence of Heterogeneity by Audience Divergence (Demographic Attributes)

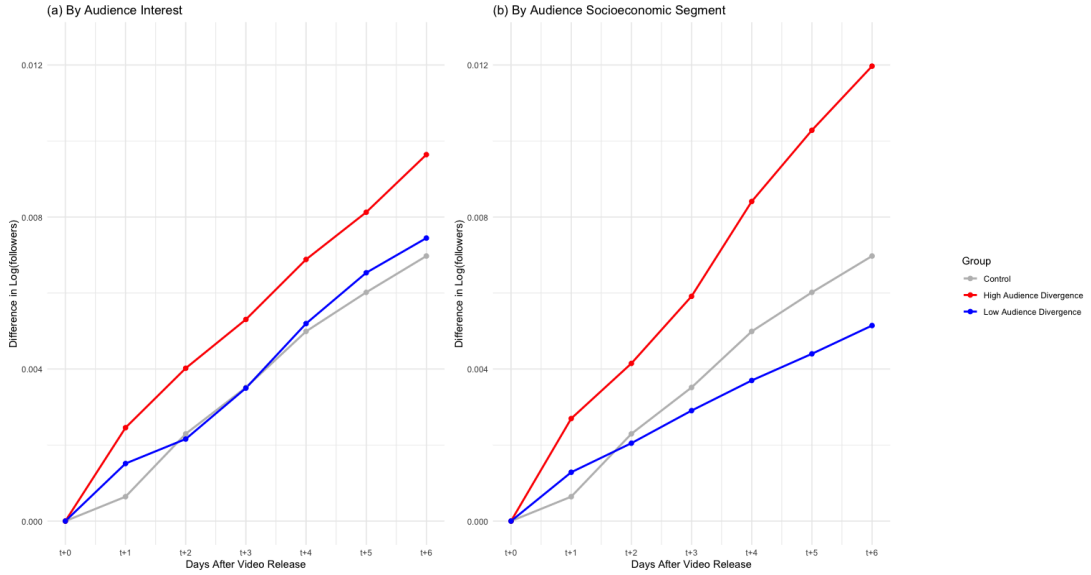


Figure 4b. Model-Free Evidence of Heterogeneity by Audience Divergence (Behavioral Attributes)

Notes: High (low) Audience Divergence Group refers to collaborations with audience divergence measure greater (lower) than the median. The Control Group (solo) includes solo instances.

Table 2. Heterogeneity by Audience Divergence (Demeaned)

	$DV: DifferenceInLog(followers)^2_{ij}$				
	Interest (1)	Socioeconomic Segment (2)	Gender (3)	Age (4)	City-tier (5)
Collaboration	0.0014*	0.0014*	0.0014*	0.0014*	0.0014*
	(0.0007)	(0.0007)	(0.0007)	(0.0007)	(0.0007)
Collaboration × $FollowerDifference_{ij}$	-0.0008***	-0.0008***	-0.0007***	-0.0007***	-0.0007***
	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)
Collaboration × $AudienceDivergence_{ij}$	0.0071**	0.0116**	0.0041	0.0034	0.0069
	(0.0033)	(0.0055)	(0.0057)	(0.0032)	(0.0044)
Controls	Included	Included	Included	Included	Included
Fixed effects	Included	Included	Included	Included	Included
Observations	1,453	1,453	1,457	1,453	1,453
R^2	0.1306	0.1340	0.1298	0.1299	0.1312

Notes. *p<0.1; **p<0.05; ***p<0.01; Estimates are rounded to four decimal digits. Clustered (at video and creator level) standard errors in parentheses.

Table 2 presents the estimates employing demeaned measures of audience divergence as a moderator on the outcome measured two days after posting the video. Here, we also control for collaboration *j*'s $FollowerDifference_{ij}$, defined as the difference in log (followers) between creator *i* and his/her collaborator. The adjustment ensures that we account for differences in audience sizes, which could otherwise confound the relationship between audience divergence and collaboration outcomes (Deshmane and Martínez-de-Albéniz, 2023).

Column (1) shows the interaction effect between collaboration and audience divergence in subject-matter interests. The coefficient is positive and statistically significant ($p<0.05$). This suggests that collaborations between creators with greater differences in audience subject-matter interests generate stronger follower growth effects. Column (2) shows a similar pattern for audience divergence in socioeconomic segments, where the interaction term is again positive and significant ($p<0.05$). Both results highlight the value of bridging social capital: collaborations are more effective when they bridge across different audience groups, rather than reinforcing the creator's existing audience base.

Columns (4) to (6) examine the moderating impact of audience divergence in demographic attributes: gender, age, and living city-tier. Across these specifications, the estimated coefficients for interaction are positive but not statistically significant. This suggests that demographic differences between collaborators' audiences play a relatively limited role in influencing audience following behavior, perhaps reflecting earlier findings on the frequent inefficacy of demographic targeting in digital advertising (e.g., Jensen et al., 2013; Boston Digital, 2024; Marketing Week, 2024). In contrast, consistent with our findings here, behavioral signals derived from users' online activities, such as interest profiles and socioeconomic indicators, have often proven more effective predictors of digital engagement (e.g., Trusove, Ma, and Jamal, 2016; Luo et al., 2014; Adamopoulus, Ghose, and Todri, 2018).

These results remain consistent when we vary the outcome window to one and three days after a video's release, demonstrating the robustness of our findings.

5.3 Collaboration Frequency

Table 3 reports the heterogeneity in collaboration effects based on creators' content strategies during the three days following the focal video. The main effect of collaboration is negative and statistically insignificant across all outcome windows (k = 3, 4, 5). This suggests that when creators do not post any new content after a collaboration, the follower growth from that collaboration largely dissipates after several days. Such a pattern aligns with a widely noted practitioner insight: extended inactivity is not encouraged in social media platforms where visibility is closely tied to frequent postings.[11]

The interaction term shows that posting solo videos within three days meaningfully amplifies the benefits of collaboration. The coefficient on *Collaboration* × *Posting Solo After* is positive and statistically significant across all specifications, indicating that posting solo content after a collaboration generates a larger follower growth impact than remaining inactive. This reflects the importance of reestablishing individual identity after a collaboration and engaging with newly acquired followers, who may still be forming their attachment to the creator. By posting solo content, creators can reinforce their personal brand identity and strengthen their relationship with their new followers.

On the other hand, the interaction *Collaboration* × *Posting Collaboration After* is positive but statistically insignificant across all windows, suggesting that releasing another collaborative videos within a three-day window after the initial one does not significantly perform better than remaining inactive.

Overall, these findings highlight the importance of after-collaboration content activities in shaping the longer-run returns to collaboration. Timely solo content, rather than additional collaborations or inactivity, provides the most effective way to sustain the momentum after the initial collaboration.

Table 3. Heterogeneity by After-Collaboration Content Strategies

[11]https://ads.tiktok.com/business/en-US/blog/tiktok-creative-ultimate-guia videode?redirected=1&acq_banner_version=73758464

	$DV: DifferenceInLog(followers)^{k}_{ij}$		
	k = 3	*k = 4*	*k = 5*
	(1)	(2)	(3)
Collaboration	-0.0006	-0.0013	-0.0015
	(0.0008)	(0.0012)	(0.0014)
Collaboration ×	0.0024**	0.0038*	0.0046*
$FollowerUpAction_{ij}$*: Posting Solo*	(0.0012)	(0.0020)	(0.0024)
Collaboration ×	0.0046	0.0052	0.0048
$FollowerUpAction_{ij}$ *: Posting Collaboration*	(0.0039)	(0.0048)	(0.0050)
Controls	Included	Included	Included
Fixed effects	Included	Included	Included
Observations	1,451	1,453	1,447
R^2	0.0992	0.0930	0.0922

Notes. *p<0.1; **p<0.05; ***p<0.01; Estimates are rounded to four decimal digits. Clustered (at video and creator level) standard errors in parentheses.

6. Robustness Check

6.1 Validating Parallel Pre-Treatment Trends in the Matched Sample

To assess the validity of our matched design, we test for parallel pre-treatment trends in the outcome for the matched sample. This also helps address concerns about reverse causality, e.g., creators expecting slower follower growth may proactively pursue collaborations to regain visibility. Using daily follower counts before the focal video posting, we compute differences in the log of follower counts over pairs of sequential days. We then estimate an event study specification (Autor, 2003), comparing daily follower changes between the treatment and control groups. As demonstrated in Equation (5), the model includes several pre-treatment period indicators (up to the prior five days), with the day just prior to posting serving as the reference period.

$$log(followers_{ij,t}) - log(followers_{ij,t-1})$$

$$= \sum_{k \neq -1} \beta_k \cdot [1(EventTime_{ij,t} = k) \cdot Collaboration_{ij}] + \gamma \cdot Controls_{ij} + \varepsilon_{ij,t} \quad (5)$$

Here, $EventTime_{ij,t} = t - T_{ij}$ captures days *k* relative to the time when creator *i* posted focal video *j*; β_k captures the average difference in daily growth between treated and control units *k* days prior to the focal video posting, as compared to the difference observed just prior to the focal video posting, i.e., *k* = -1.

Finally, $Collaboration_{ij}$ indicates whether the focal video, *j*, from creator *i*, belongs to the treatment group, and $Controls_{ij}$ once again captures our vector of matching covariates.

Table 4 presents the results of this estimation. All coefficients (β_k) are statistically insignificant, suggesting that creators associated with the matched treatment and control groups exhibit similar daily follower growth trends before focal video posting.

Table 4. Estimated Coefficients of the Pre-Treatment Dummies

	Coefficients
Collaboration × *k = -5*	0.000264 (0.000202)
Collaboration × *k = -4*	0.000227 (0.000214)
Collaboration × *k = -3*	0.000194 (0.000214)
Collaboration × *k = -2*	0.000137 (0.000167)
Collaboration × *k = -1*	_
Controls	Included
Fixed effects	Included
Observations	7,160
R^2	0.0327

Notes. *p<0.1; **p<0.05; ***p<0.01; Estimates are rounded to three significant digits. Clustered (at video and creator level) standard errors in parentheses.

7. Discussion

Audience growth is increasingly a challenge for social media creators, and collaborations have emerged as a popular strategy for expanding reach. Yet, the optimal way to collaborate remains underexplored in this domain. Our findings offer several actionable insights for creators on how to optimize their collaboration strategies. First, creators should prioritize collaborators whose audiences differ substantially from their own—especially along behavioral dimensions such as interests and socioeconomic segments, rather than demographic attributes like age or gender. Second, creators should follow a collaboration with timely solo posts to reinforce their individual identity and strengthen ties with newly acquired followers. Finally, creators should avoid launching another collaboration too quickly; posting an additional collaborative video within a few days can dilute the gains from the initial collaboration. Together, these insights highlight how thoughtful partner selection and strategic after-collaboration content activities can meaningfully enhance collaboration effectiveness.

Beyond individual creators, our findings also carry important implications for platforms and agencies that recommend collaboration partners (e.g., Kit.com, TikTok Creator Marketplace). Current matching systems often prioritize creator-centric metrics like follower count, creator genre, or specialized content areas. However, our results highlight the importance of considering collaborators' audience characteristics, such as interests and city tiers. Considering these additional factors, platforms and collaborator matching intermediaries can help creators forge effective collaborations that lead to maximal follower growth.

Our research contributes to the collaboration literature by extending it to the context of social media—an environment that differs meaningfully from traditional media domains such as branding, music, and performance arts (e.g., Cao and Sorescu, 2013; Deshmane and Martínez-de-Albéniz, 2023; Uzzi and Spiro, 2005). In particular, we reframe partner selection strategy within the interactive dynamics of social media content creation. Prior work has largely focused on trade-offs in selecting collaboration partners with similar or different creator characteristics, emphasizing attributes such as brand image, social status, or network position. We introduce a new dimension to partner selection: the composition of creators' audiences. Social media embeds creators in continuous, bidirectional engagement with their audiences, who actively shape both content and collaboration. By shifting the focus from collaborator-level attributes to audience-level divergence, we highlight how audience differences can moderate the impact of collaboration on follower growth.

Furthermore, our study introduces social capital theory as a fresh theoretical lens for examining collaboration choices on social media. Social capital has long been recognized as a key driver of value creation in traditional organizational and community contexts (Coleman, 1988; Nahapiet and Ghoshal, 1998; Putnam, 2002). Recently, many works in the digital area have begun to extend the concept into settings such as online community leadership (Forderer and Burtch, 2025), music production (Deshmane and Martínez-de-Albéniz, 2023), and user satisfaction (Sun et al., 2012). We build on this stream by applying social capital theory to social media creators, offering a conceptual framework to explain how collaboration strategies shape value creation from partnership.

Lastly, we contribute to the research on influencer marketing by investigating how creators' joint efforts shape marketing outcomes. While prior studies in this stream predominantly emphasize individual efforts on consumer engagement (Hughes, Swaminathan, and Brook, 2019; Wiles, Bleier, and Edeling, 2022; Leung et al., 2022), influencer attitude (De Veirman, Cauberghe, and Hudders, 2017; De Veirman and Hudders, 2020; Zheng and Zhang, 2024), purchase propensities (Lee and Eastin, 2020), and product sales (Bharadwaj et al., 2022), few investigate collaborative influencer actions. One exception is Gu et al. (2024), which explores the combinatorial effect of employing big and small influencers on product sales. However, such joint action does not involve direct collaboration between influencers or any content

co-creation. Our study, by contrast, considers when influencers co-create content and examines the effectiveness of various collaboration strategies in driving follower growth.

This research is not without limitations, which also open avenues for future inquiry. First, because of platform constraints on data collection for small-scale creators, our dataset underrepresents nano-creators. Readers should therefore exercise caution when generalizing our findings to this segment, as nano-creators may exhibit different collaboration dynamics. Second, our analysis of partner selection and after-collaboration actions is conditional on the content produced. Our measure of video content is imperfect, as it relies on 16-dimensional embeddings derived solely from video transcripts, capturing only textual information and omitting the visual and audio components that are central to short-form video. A natural next step would be to develop richer content representations that integrate all modalities (text, audio, and visuals) to more fully characterize video content. Another promising direction would be to examine how specific creative choices within collaborative videos—such as stylistic elements, visual aesthetics, editing patterns, or the balance of screen time between collaborators—influence audience responses and follower growth.

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APPENDICES

Appendix 1a. Summary Statistics of Audience Characteristics (Segment Profile).

	Mean	SD	Min.	Max.
Audience Segement Profile				
Urban middle class	0.113	0.065	0.002	0.449
Urban white-collar	0.091	0.075	0.001	0.463
Urban blue-collar	0.099	0.054	0.007	0.430
Urban senior	0.162	0.128	0.003	0.760
Rural junior	0.155	0.077	0.003	0.735
Rural senior	0.191	0.114	0.003	0.764
Generation Z	0.136	0.151	0.002	0.694
Stylish mom	0.053	0.042	0.000	0.309

Notes. Statistics are rounded to the third decimal digit.

Appendix 1b. Summary Statistics of Audience Characteristics (Interest Profile).

	Mean	SD	Min.	Max.
Audience Interest Profile				
Vlogging	0.076	0.021	0.044	0.173
Countryside	0.031	0.029	0.000	0.096
Life Sharing	0.063	0.018	0.035	0.128
Sports	0.025	0.022	0.000	0.080
Music	0.011	0.019	0.000	0.065
Food	0.055	0.009	0.000	0.086
Fashion	0.043	0.018	0.000	0.100
Home Fashion	0.022	0.022	0.000	0.059
Politics	0.022	0.022	0.000	0.062
Car	0.007	0.016	0.000	0.059
TV Drama	0.013	0.020	0.000	0.076
Education	0.018	0.039	0.000	0.140
Entertainment	0.014	0.032	0.000	0.103
News	0.013	0.028	0.000	0.086
Talent	0.009	0.021	0.000	0.075
Celebrity	0.008	0.017	0.000	0.074
Game	0.010	0.021	0.000	0.078
Animation	0.010	0.020	0.000	0.075
Relationship	0.000	0.002	0.000	0.062
Dance	0.000	0.004	0.000	0.059
Travel	0.000	0.004	0.000	0.049
Culture	0.000	0.004	0.000	0.059
Film	0.002	0.009	0.000	0.047
Technology	0.002	0.008	0.000	0.057
Medical Health	0.001	0.008	0.000	0.055
Campus	0.000	0.000	0.000	0.000
Finance	0.000	0.007	0.000	0.059
Science	0.000	0.000	0.000	0.004
Parent	0.000	0.006	0.000	0.123
Animal Plant	0.000	0.000	0.000	0.000
Uncategorized	0.000	0.000	0.000	0.000

Notes. Statistics are rounded to the third decimal digits.

Appendix 1c. Summary Statistics of Audience Characteristics (Demographic Profile).

	Mean	SD	Min.	Max.
Audience Gender Profile				
Male	0.567	0.213	0.043	0.985
Female	0.433	0.213	0.015	0.957
Audience Age Profile				
18-23	0.115	0.154	0.002	0.797
24-30	0.137	0.094	0.007	0.513
31-40	0.332	0.131	0.022	0.704
41-50	0.193	0.096	0.013	0.502
>50	0.222	0.186	0.002	0.896
Audience City Tier Profile				
First	0.078	0.053	0.002	0.793
New First	0.181	0.081	0.002	0.880
Second	0.174	0.074	0.002	0.881
Third	0.235	0.102	0.010	0.862
Forth	0.193	0.104	0.001	0.891
Fifth	0.130	0.120	0.003	0.930
Sixth and Below	0.010	0.019	0.000	0.794
Special Administration	0.000	0.000	0.000	0.002

Notes. Statistics are rounded to the third decimal digits.

Appendix 2a. Balance Test of Standardized Difference (in Means) for Creator Characteristics

	Before Matching			After Matching		
	Collab Mean	Solo Mean	Std. Diff.	Collab Mean	Solo Mean	Std. Diff.
log(FollowersBeforeCollaboration)	11.844	12.805	-0.558	11.774	11.681	0.054
log(Career Length)	21.261	21.261	0.132	7.385	7.393	-0.016
Average Engagement Before	16003	60232	-0.560	15010	16160	-0.015
Collaboration Frequency Before	0.135	0.000	0.644	0.005	0.001	0.020
Num. of Solo 1 Day Before	0.449	0.628	-0.219	0.545	0.526	0.024
Num. of Solo 2 Days Before	0.490	0.721	-0.299	0.541	0.529	0.016
Num. of Solo 3 Days Before	0.453	0.685	-0.331	0.488	0.471	0.024
Num. of Solo 4 Days Before	0.456	0.706	-0.358	0.521	0.495	0.037
Num. of Solo 5 Days Before	0.462	0.679	-0.299	0.507	0.519	-0.017
Num. of Collab 1 Day Before	0.078	0.018	0.216	0.015	0.022	-0.020
Num. of Collab 2 Days Before	0.075	0.024	0.192	0.015	0.015	0.000
Num. of Collab 3 Days Before	0.055	0.022	0.144	0.025	0.026	-0.006
Num. of Collab 4 Days Before	0.058	0.024	0.143	0.027	0.030	-0.012
Num. of Collab 5 Days Before	0.045	0.022	0.110	0.025	0.016	0.040

Notes. Means and differences are rounded to the third decimal digits. Collab., treatment; Solo., control;

Appendix 2b. Balance Test of Standardized Difference (in Means) for Audience Characteristics

	Before Matching			After Matching		
	Collab Mean	Solo Mean	Std. Diff.	Collab Mean	Solo Mean	Std. Diff.
Audience PC1	-0.353	-0.612	0.107	-0.543	-0.588	0.019
Audience PC2	-0.727	0.030	-0.330	-0.612	-0.795	0.080
Audience PC3	0.317	-0.245	0.320	0.269	0.259	0.006
Audience PC4	0.063	0.037	0.015	0.002	-0.049	0.030
Audience PC5	0.208	-0.061	0.149	0.220	0.189	0.017
Audience PC6	-0.048	-0.002	-0.039	-0.067	-0.035	-0.027
Audience PC7	0.001	-0.104	0.085	-0.021	0.014	-0.029
Audience PC8	-0.035	0.005	-0.027	-0.014	-0.057	0.029
Audience PC9	0.135	0.004	0.105	-0.070	0.050	0.016
Audience PC10	-0.028	-0.022	-0.004	-0.031	0.012	-0.035
Audience PC11	-0.096	0.048	-0.132	-0.132	-0.194	0.056
Audience PC12	0.071	-0.017	0.078	0.062	0.055	0.007
Audience PC13	0.068	-0.032	0.088	0.013	0.078	-0.057
Audience PC14	0.145	-0.041	0.164	0.110	0.101	0.007
Audience PC15	-0.114	0.024	-0.109	-0.168	-0.220	-0.041
Audience PC16	0.008	-0.005	0.044	0.016	0.019	-0.009
Audience PC17	0.067	0.003	0.056	0.048	0.004	0.039
Audience PC18	-0.058	-0.004	-0.053	-0.072	-0.068	0.003
Audience PC19	-0.027	0.002	-0.030	0.014	-0.025	0.039
Audience PC20	-0.032	0.016	-0.031	-0.023	-0.067	0.034
Audience PC21	0.047	-0.005	0.053	0.032	0.007	0.026
Audience PC22	0.136	-0.023	0.170	0.103	0.120	-0.018
Audience PC23	-0.126	0.039	-0.177	-0.103	0.090	-0.014
Audience PC24	-0.059	0.052	-0.140	-0.010	-0.131	0.039
Audience PC25	0.094	0.006	0.107	0.109	0.093	0.020
Audience PC26	0.041	0.008	0.042	0.036	0.021	0.018

Notes. Means and differences are rounded to the third decimal digits. Collab., treatment; Solo., control;

Appendix 2c. Balance Test of Standardized Difference (in Means) for Video Characteristics

	Before Matching			After Matching		
	Collab Mean	Solo Mean	Std. Diff.	Collab Mean	Solo Mean	Std. Diff.
Content Embedding 1	0.071	0.075	-0.044	0.071	0.071	-0.003
Content Embedding 2	-0.111	-0.114	0.030	-0.109	-0.112	0.029
Content Embedding 3	0.060	0.060	-0.014	0.060	0.062	-0.056
Content Embedding 4	-0.010	-0.014	0.031	-0.011	-0.010	-0.004
Content Embedding 5	0.007	0.003	0.037	0.008	0.018	-0.043
Content Embedding 6	-0.280	-0.282	0.018	-0.281	-0.280	-0.002
Content Embedding 7	0.158	0.160	-0.016	0.156	0.154	0.023
Content Embedding 8	0.543	0.540	0.062	0.543	0.543	0.003
Content Embedding 9	-0.309	-0.308	-0.014	-0.309	-0.308	-0.011
Content Embedding 10	0.106	0.108	-0.018	-0.342	-0.344	0.026
Content Embedding 11	-0.344	-0.345	0.016	-0.348	-0.347	-0.010
Content Embedding 12	-0.179	-0.178	-0.008	-0.183	-0.187	0.043
Content Embedding 13	-0.103	-0.104	0.010	-0.103	-0.104	0.018
Content Embedding 14	0.085	0.084	0.013	0.087	0.086	0.009
Content Embedding 15	-0.308	-0.312	0.034	-0.308	-0.301	-0.052
Content Embedding 16	0.269	0.272	-0.027	0.269	0.272	-0.027

Notes. Means and differences are rounded to the third decimal digits. Collab., treatment; Solo., control;